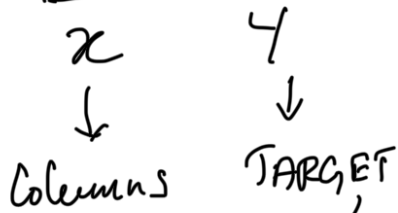


① SUPERVISED

Predicts - 'y'

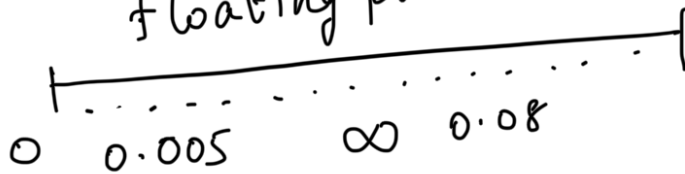


x's & y's

We are here!!

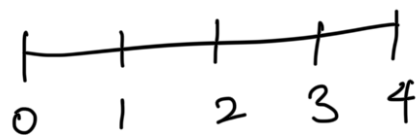
REGRESSION

1. If my target => Continuous floating pt values

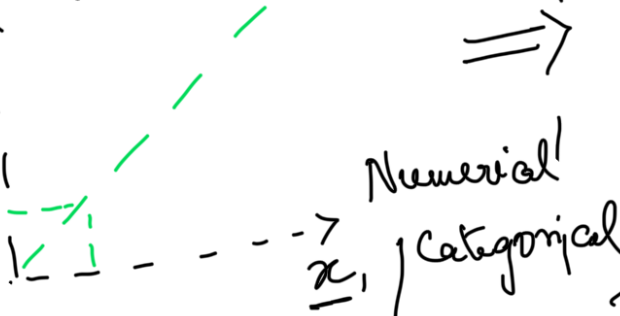


CLASSIFICATION

1. If I go on to predict discrete values



① Linear Regression



Take 'n' x values map them to a regression line

input: x values
Output: Continuous Nos.

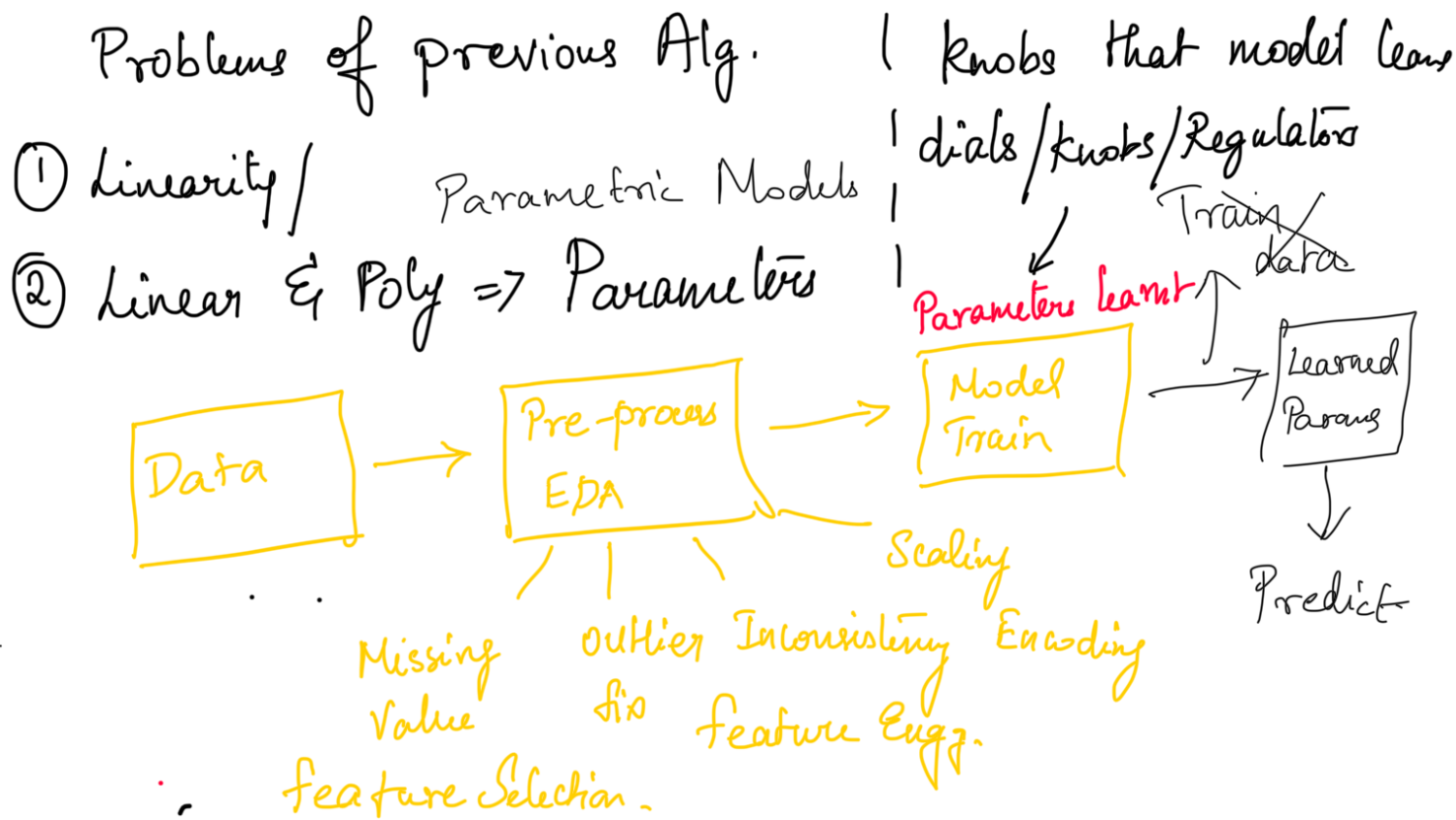
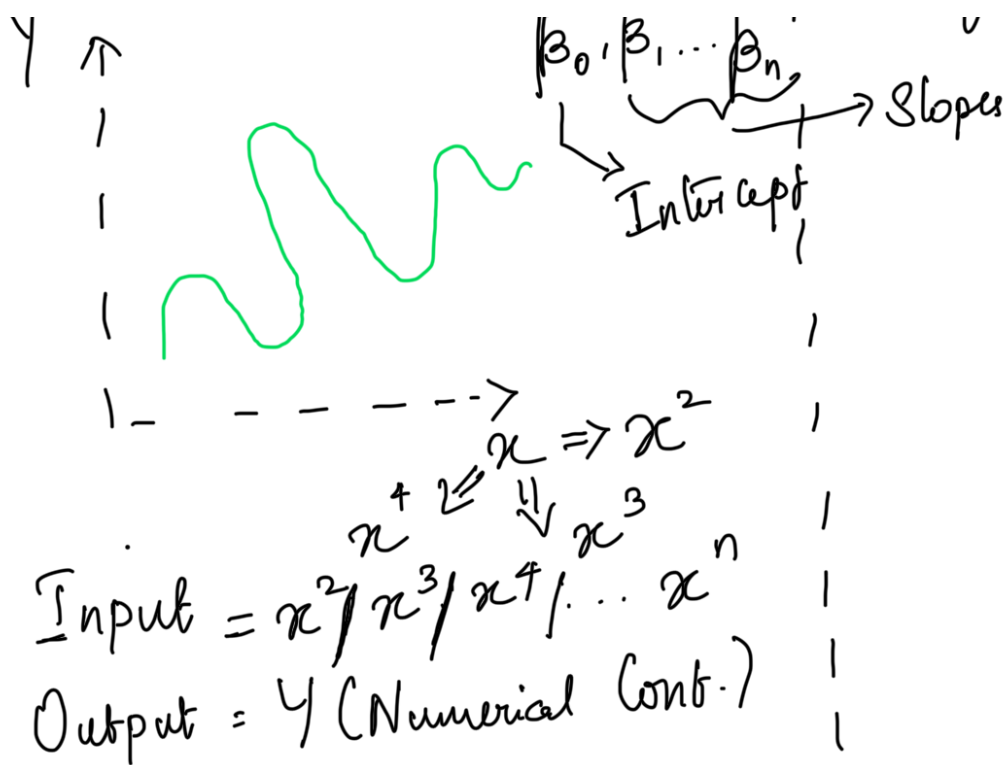
② Logistic Regression

Take h.R outputs map them probabilities

input: x values
Output: Sigmoid Prob.

③ Polynomial Regression

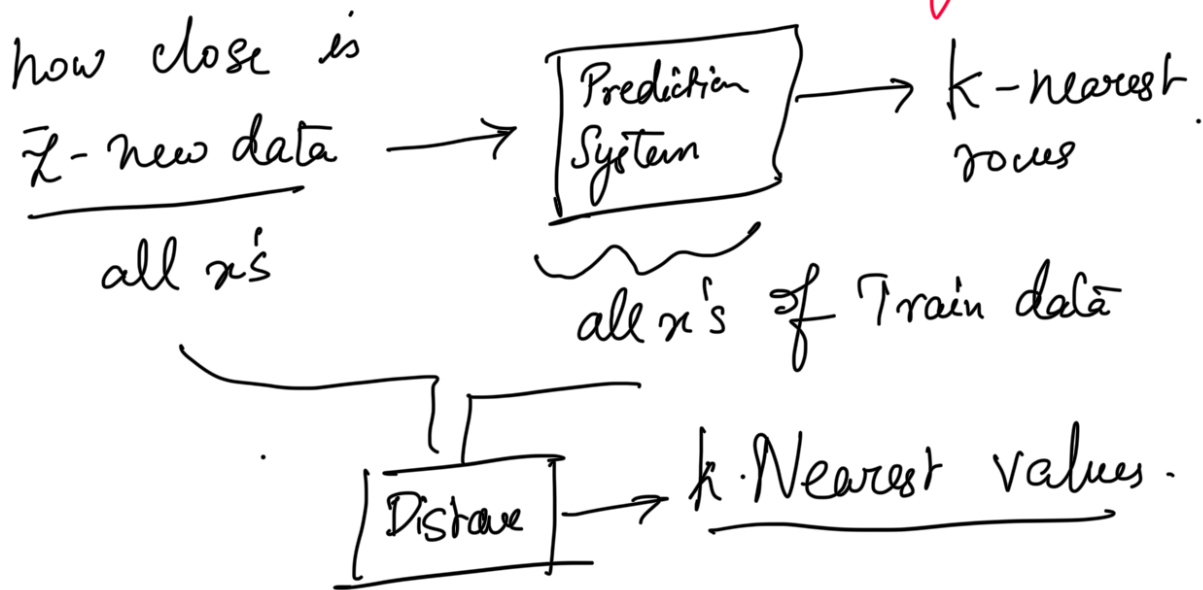
④ Polynomial - Logistic



KNN $\Rightarrow$ Make no assumptions about Relationships
 $\Rightarrow$ It does not have learnable dials/knobs

Manhattan & cosine Distance

$\Rightarrow$ Euclidean Distance | Minousky.



Regression :-

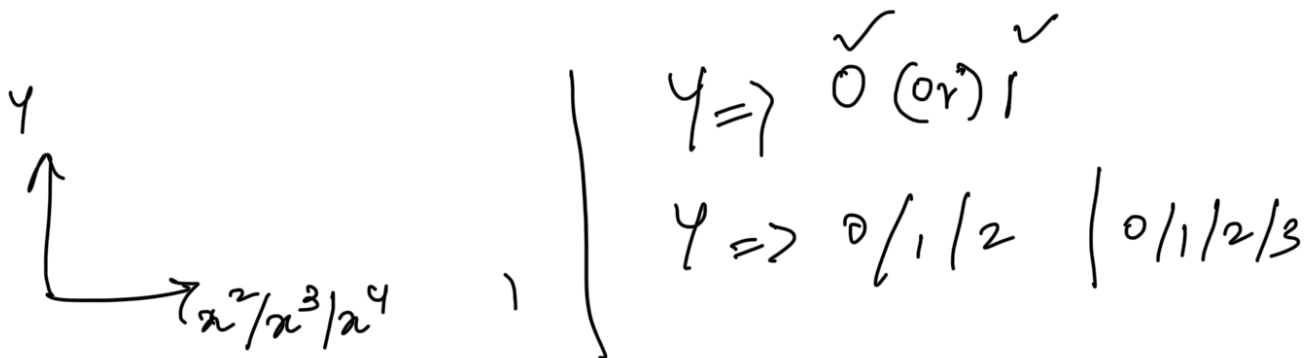
k -averaged
& output
a continuous
No.

Classification :-

k -MODE
& output
discrete
decision.

Problems:-

① Lazy learner $\Rightarrow$ All the zones in Memory.
Predictions are going to change on the fly.



INPUT = X variables.

Output = low risk - / medium / high

One vs Rest $\Rightarrow$ 3 Models / 'n' class Models

Model A $\Rightarrow$ low risk vs { Medium , High }

Model B $\Rightarrow$ Medium risk v { Low , High }

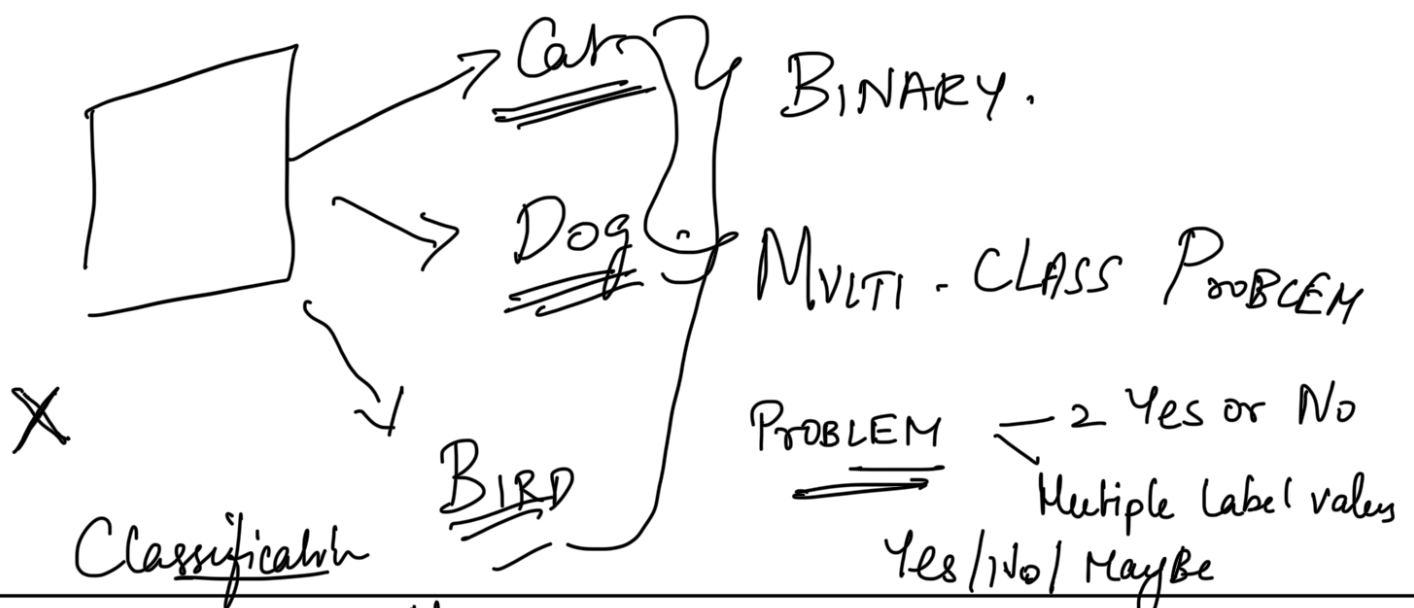
Model C $\Rightarrow$ High risk v { Low , Medio }

① Interpretation

$\uparrow$ Each Model tries to Isolate

Specific class

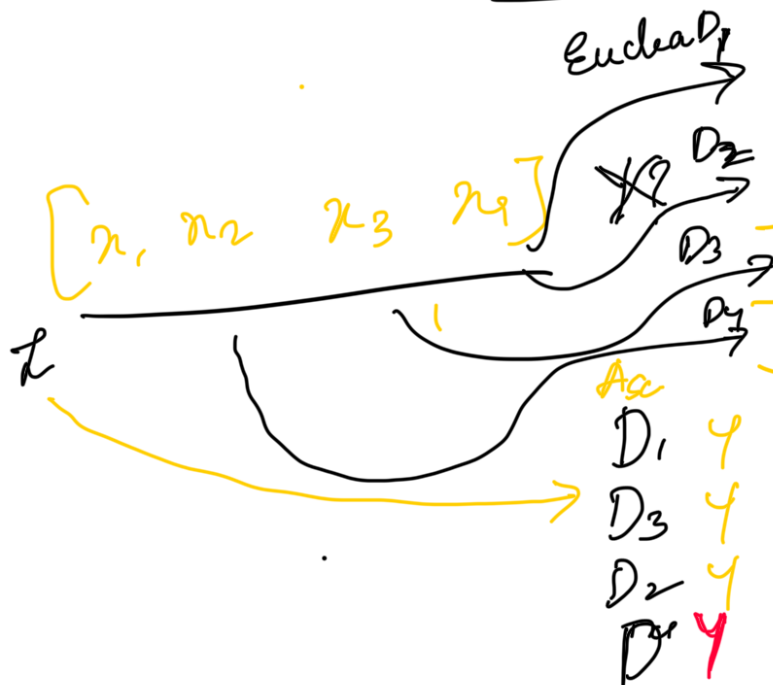
$K \Rightarrow k = [k_1, k_2, k_3, k_4 \dots k_c]$



KNN $\times$ - Numerical / Categorical | DT - R | BINARY



• → No Category:



Mode of k Shortest distance will be final decision.

$$\begin{matrix} 4 \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{matrix} \quad \left. \begin{matrix} 4 \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix} \right\} \quad \frac{4}{12} = \frac{1}{3}$$

$$\begin{matrix} 4 \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{matrix} \quad \left. \begin{matrix} 4 \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix} \right\} \quad \frac{4}{12} = \frac{1}{3}$$

$$- \sum_{i=1}^k P(x_i) \log P(x_i)$$

$$\rightarrow -3 \times \left(\frac{1}{3} \log \left(\frac{1}{3} \right) \right)$$

$$\left. \begin{matrix} 1 \\ 2 \\ 2 \\ 2 \\ 2 \end{matrix} \right\} 4/12 = 1/3 \quad \underline{\underline{1.58}}$$

Maximum possible Temp. / $H_{\max} = \underline{\underline{1.585}}$
 $P_0 = P_1 = P_2 = 1/3$

$$0, 0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3$$

$$P_0 = P_1 = P_2 = P_3 = 1/4 = 0.25$$

3
2

$$= -4 \left(\frac{1}{4} \log_2 \left(\frac{1}{4} \right) \right)$$

$$H(S) = -\log_2(2^{-2}) = -(-2) = \underline{\underline{2.0}}$$

When a decision Tree has k -class

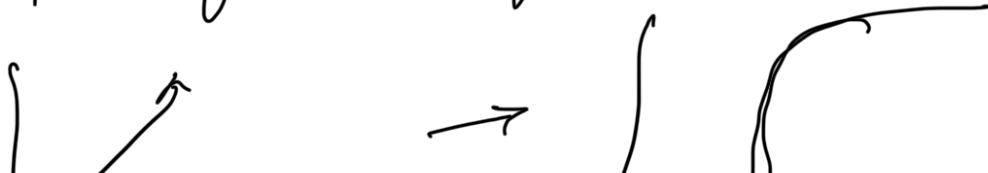
$$\underline{\log_2(k)}$$

$$\max(H(S)) = \log_2(k) - \log_2(8)$$

$\log_2 2^3$

$\approx \log_2$

Logistic Regression: $\Rightarrow$ Engineering fix
 Output of Linear Regression $\rightarrow$ Probs



$k = 3$ classes

Outputs



$$b_0 + b_1 x$$

$$\log_2(10)$$

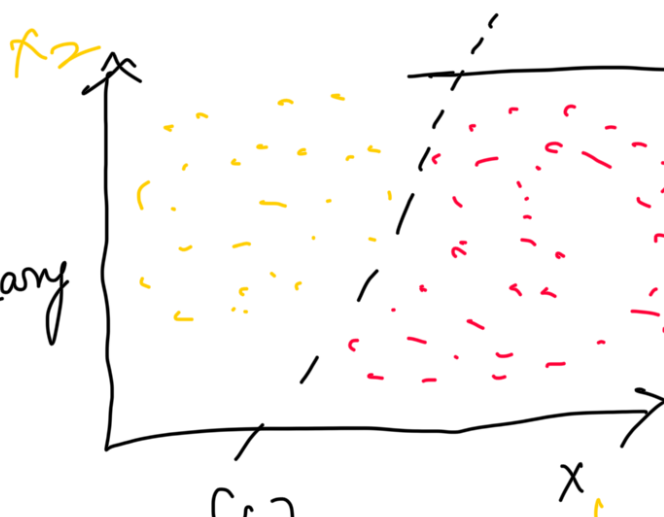
Engineering fit

Confusion

Intuitions

Binary

Not on logistic part



Row

hour

Real.

One-hot [y]

Row	hour	Real.	One-hot [y]
1	2.0	low	[1 0 0]
2	2.0	low	[1 0 0]
3	2.0	low	[1 0 0]
4	2.0	low	[1 0 0]
5	6.0	Medium	[0 1 0]
6	6.0	Medium	[0 1 0]
7	6.0	Medium	[0 1 0]
8	6.0	Medium	[0 1 0]
9	10.0	High	[0 0 1]
10	10.0	High	[0 0 1]
11	10.0	High	[0 0 1]
12	10.0	High	[0 0 1]

3 classes $N=12$

$k=3$ classes - Softmax Regression =

We need more than 1 line for multiclass Classification / So Linear Regression itself is reworked for Multi-class Classification

$$1 \Rightarrow \beta_0 + \beta_1 x \Rightarrow 0 \quad | \quad 1$$

If I have 'k' classes.

$$y_k = W_k x + b \quad | \quad \beta_0 + \beta_1 x$$

Step 1

3 - classes $\Rightarrow$ 3 - Regression line Eqs

$\rightarrow$ Class 0 $\Rightarrow$ low Risk $\Rightarrow [1, 0, 0] =$ Putting
 $\rightarrow$ Class 1 $\Rightarrow$ Medium Risk $\Rightarrow [0, 1, 0] =$ Classes
 $\rightarrow$ Class 2 $\Rightarrow$ ^{High} Risk $\Rightarrow [0, 0, 1] =$ as
 Hot vector

Class 0	Class 1	Class 2
$y_0 = \beta_0 + \beta_1 x$	$y_1 = \beta_0 + \beta_1 x$	$y_2 = \beta_0 + \beta_1 x$
$\beta_0 \rightarrow \beta_1$	$\beta_0 \rightarrow \beta_1$	

$dL/d\beta_0, dL/d\beta_1$ $\downarrow$ Inference new loan ask. 1 - 3.131 Labels

Classes		
Class 0	Class 1	Class 2
$\beta_0 = 4.5$ $\beta_1 = -1.20$ $z_0 = (-1.20 \times 3.31) + 4.5$ $= +0.528$	$\beta_0 = 0.80$ $\beta_1 = 0.25$ $z_1 = (0.25 \times 3.31) + 0.80 = \underline{\underline{1.62}}$	$\beta_0 = -3.20$ $\beta_1 = 1.60$ $z_2 = \underline{\underline{0.441}}$

Converting Contin values into probab
 (Sum Total probability of all class add up to 1) $= \frac{1}{1+e^{x_2}}$

SOFTMAX $\hat{p}_k = \frac{e^{z_k}}{e^{z_0} + e^{z_1} + e^{z_2}}$ (Standard) 0-1

$$e^{0.528} + e^{1.628} + e^{0.441} = 1.69 + 5.09 + 1.5 = \underline{\underline{8.34}}$$

low $\frac{1.69}{8.34} = 0.20$ High $= \frac{1.5}{8.3} = 0.18$

Med $= \frac{5.09}{8.34} = 0.61$

$[0.2, 0.61, 0.18]$

Not in Supr.

Multi-output
segmentation

y_1, y_2, y_3
↑ ↓
~~~~~

output Medium Risk

## AGENDA

- ① Classification?
- ② Binary vs Multiclass.
- ⇒ ③ loss vs Metric.
- ⇒ ④ CONFUSION MATRIX